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Fully automatic tracking of native glenohumeral kinematics from stereo-radiography

William Burton [a,*], Ignacio Rivero Crespo [a], Thor Andreassen [a], Moira Pryhoda [a], Andrew Jensen [b], Casey Myers [a], Kevin Shelburne [a], Scott Banks [b], Paul Rullkoetter [a]

[a] Center for Orthopaedic Biomechanics, University of Denver, 2155 E. Wesley Ave., Denver, CO, 80210, USA
[b] Department of Mechanical and Aerospace Engineering, University of Florida, 939 Center Dr., Gainesville, FL, 32611, USA

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ABSTRACT

The current work introduces a system for fully automatic tracking of native glenohumeral kinematics in stereo-radiography sequences. The proposed method first applies convolutional neural networks to obtain segmentation and semantic key point predictions in biplanar radiograph frames. Preliminary bone pose estimates are computed by solving a non-convex optimization problem with semidefinite relaxations to register digitized bone landmarks to semantic key points. Initial poses are then refined by registering computed tomography-based digitally reconstructed radiographs to captured scenes, which are masked by segmentation maps to isolate the shoulder joint. A particular neural net architecture which exploits subject-specific geometry is also introduced to improve segmentation predictions and increase robustness of subsequent pose estimates. The method is evaluated by comparing predicted glenohumeral kinematics to manually tracked values from 17 trials capturing 4 dynamic activities. Median orientation differences between predicted and ground truth poses were 1.7° and 8.6° for the scapula and humerus, respectively. Joint-level kinematics differences were less than 2° in 65%, 13%, and 63% of frames for XYZ orientation DoFs based on Euler angle decompositions. Automation of kinematic tracking can increase scalability of tracking workflows in research, clinical, or surgical applications.

1. Introduction

Model-image registration enables precise tracking of *in vivo* joint-level kinematics in radiograph sequences in biomechanics and orthopaedics methods [1–5]. Also referred to as 2D-3D registration, this procedure involves estimating 6-degree of freedom (6-DoF) poses of objects observed in radiographs by comparing captured frames to dynamically rendered images of digitized models. Scenes are iteratively generated as a function of pose to find configurations which replicate the appearances and shapes of target objects in original radiographs. Rendered images are instantiated as digitally reconstructed radiographs (DRRs) or surface model perspective projections [6–8], acquired from computed tomography (CT) scans [9,10] or magnetic resonance imaging (MRI) [11–13].

Individual object poses estimated with model-image registration are represented with respect to semantically relevant coordinate systems and aggregated to recover joint-level kinematics [14–16]. Model-image registration has been successfully applied to compute kinematics of the native or implanted knee [1,2,4,17–21], shoulder [3,13,22–26], spine [27,28], and other joints [29,30] from single plane [17] or biplanar systems [2,4]. In particular, previous studies evaluated glenohumeral (GH) kinematics differences between dominant and non-dominant joints [31], or compared healthy to injured or implanted shoulders [32–35].

Kinematic tracking in radiographs has most commonly been performed manually or semi-automatically. Manual tracking requires that trained observers iteratively specify multiple poses until rendered images of target objects are sufficiently similar to captured radiographs. In contrast, semi-automatic methods compare radiograph frames to rendered images using scalar-valued objective functions, and invoke numerical optimization to minimize such objectives as a function of pose. A wide range of objective functions [36–38] and optimization methods [29,39,40] have been employed in prior work, and have contributed a degree of automation to the tracking process. However, a stipulation of semi-automatic methods is that object poses must be carefully initialized within just a few millimeters (mm) or degrees of the true pose to cope with chaotic loss landscapes and expensive rendering operations [20,40]. Pose iterates are also susceptible to settling in undesirable local minima, and this may preclude the use of pose

* Corresponding author.
E-mail address: will.burton@du.edu (W. Burton).

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solutions from adjacent time steps for initializing new frames. As a result, manual and semi-automatic tracking methods have been noted to require multiple hours or days of human effort to fully process a single trial [39]. These drawbacks limit scalability of precise tracking workflows in research and clinical settings.

The current work introduces a system for fully automatic tracking of native joint kinematics in high-speed stereo-radiography (HSSR) sequences [41]. The proposed method consists of three distinct computational steps. Radiographs are first annotated with segmentation maps and semantic key points using convolutional neural networks (CNNs). Next, initial bone pose estimates are computed using subject-specific surface landmarks and CNN-predicted key points. Initial poses are then refined by registering CT-based DRRs to masked radiographs. A particular CNN architecture is also proposed which exploits subject-specific surface models for improved radiograph annotation; and a biplanar Perspective-n-Point (PnP) formulation is solved using semidefinite relaxations to obtain reliable, preliminary pose estimates. The introduced approach is applied to track native shoulder kinematics, in which observed radiographs are characterized by low contrast and heavy occlusions by surrounding tissue and bone. Predicted GH kinematics are evaluated by comparing to manually tracked values from 17 trials capturing 4 distinct activities performed by living individuals. Observed differences routinely within 2.0° for 2 out of 3 joint-level DoFs demonstrate the viability of end-to-end automation of kinematic tracking in stereo-radiography sequences.

2. Related work

2.1. Model-image registration

Early methods for accelerating model-image registration utilized semi-automatic frameworks. These approaches applied numerical optimization algorithms to minimize loss functions which quantify similarities between radiograph frames and dynamically rendered images. Loss functions have also been augmented with regularization terms [42] or simulated contact and ligament force constraints [29,43] to mitigate spurious local minima known to exist in the induced loss landscapes. Otake et al. [44] investigated the use of Covariance Matrix Adaptation-Evolutionary Strategies (CMA-ES) to register the proximal femur with CT-based DRRs. An intensity-based similarity metric was computed at multiple scales using progressively increasing radiograph and DRR resolutions. Mahfouz et al. [20] combined simulated annealing and Nelder–Mead simplex to register total knee arthroplasty (TKA) components by accounting for intensity and contour differences between projection maps and radiographs. Flood and Banks [8] introduced a global optimization algorithm based on hierarchical division of search space hyper-rectangles to acquire TKA kinematics from single plane fluoroscopy. Implant component poses were estimated by penalizing differences between post-processed radiograph contours and surface mesh projection maps. These methods utilized gradient-free optimization or relied on numerical gradients. More recent investigations have sought to develop methods based on analytical gradients [45,46]. Gopalakrishnan and Golland [47] implemented DRR rendering as a series of backpropagatable tensor operations to compute analytical gradients of a similarity metric with respect to scene pose. An alternative approach used neural nets to regress to pose from intensity residual maps constructed from rendered implants and radiograph frames [48]. In contrast to these previous investigations, the current work develops methods for fully automatic bone pose estimation.

Fully automatic tracking methods have recently been introduced to supplement optimization-based pose refinement with automatic initial pose estimates [49–51]. Grupp et al. [42] investigated fully automatic tracking of the pelvis and proximal femur in single plane radiography by applying CNNs to predict 2D landmark heat maps and segmentation. Landmarks enabled PnP solutions for initial pose estimates while segmentation maps were utilized for additional DRR-based pose refinement. Burton et al. [52] implemented a similar approach to track native knee kinematics in stereo-radiography. Initial pose estimates were obtained from a biplanar PnP formulation solved with block coordinate descent, and pose refinement was performed by registering surface models to masked radiographs. Gu et al. [53] proposed a CNN architecture to predict initial poses from encoded radiographs and DRRs. Recently, Jensen et al. [54] predicted preliminary implant poses in single plane radiography by matching CNN-segmented implant contours to predefined libraries of normalized Fourier descriptors (NFD) [17]. Poses were then refined using global optimization to estimate precise TKA kinematics. These previous investigations adopted a multi-stage framework characterized by three distinct steps including preliminary image annotation with machine learning, initial pose estimation, and pose refinement. The current work also adheres to this multi-stage framework but builds upon previous investigations in two key ways. A CNN architecture is proposed which leverages subject-specific geometric data to inform segmentation predictions, and a semidefinite relaxation strategy is implemented to obtain globally optimal preliminary pose estimates from a non-convex biplanar PnP formulation. The current investigation also reports results for automatic tracking of native GH kinematics, a challenging task in which radiographs are characterized by low contrast and significant occlusions. Results for this target application have been so far unreported.

2.2. Deep learning for computer vision

The emergence of large-scale CNNs has prompted development of robust machine learning (ML) models that perform a wide range of computer vision tasks. Two prevalent tasks include automatic segmentation [55–57] and semantic key point prediction [58–60]. Automatic segmentation is characterized by pixel-wise annotation of query images to delineate object boundaries while semantic key point prediction involves annotating scenes with 2D landmarks of interest. ML models have been applied to annotate radiograph frames for various purposes with promising results [52,54,61–64].

While CNN architectures have classically been composed of affine transformations and non-linear activations [65–67], recent work has also aimed to augment models with a wider range of operators for enhanced expressivity [68,69]. A precondition for these introduced operators is that they should be analytically differentiable to allow backpropagation and end-to-end training. Numerical optimization as an embedded operator has been identified as one candidate for improving expressivity of ML models [70–73]. Specifically, computing analytical gradients of optimization *solutions* with respect to learned input data has enabled solving numerical optimization problems during an architecture's forward pass, while also allowing backpropagation through optimization modules. Gradients have been derived by solving systems of equations based on Karush–Kuhn–Tucker (KKT) conditions, or with the Implicit Function Theorem [74,75].

A particular instantiation of this emerging class of differentiable modules is backpropagatable PnP, in which gradients of optimized poses can be computed with respect to input data [76,77]. Differentiable rendering has also been introduced as a backpropagatable module that can augment ML architectures. While the rasterization step of conventional computer graphics prevents computing analytical gradients, differentiable reformulations have enabled backpropagation through rendered images with respect to scene parameters, including pose [78–81]. As will be shown, the current work adopts these advanced differentiable modules as part of a CNN architecture which uses subject-specific geometric data to inform radiograph annotations.

2.3. Pose estimation from semantic key points

The PnP problem aims to recover the pose of a 3D object observed by a calibrated camera based on 3D landmarks from the object's digitized model and corresponding 2D projections on the image plane [82–

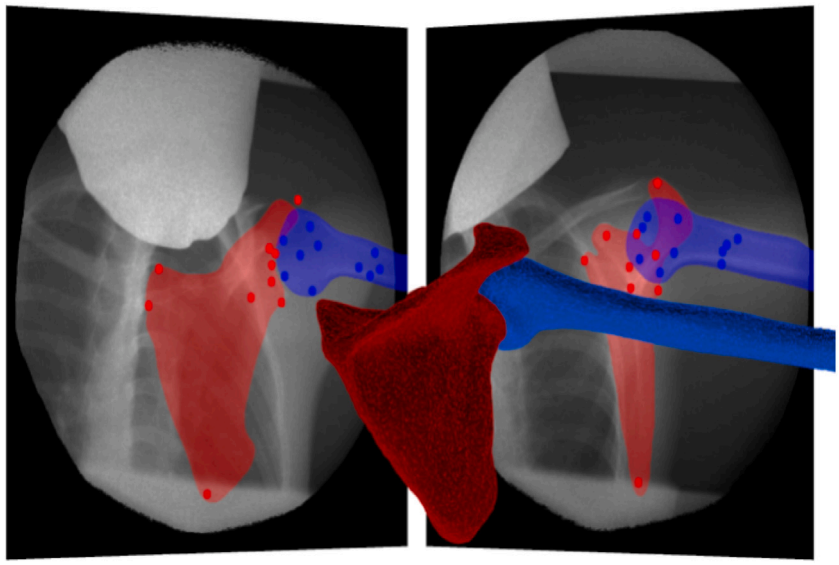

Fig. 1. Method for automatic tracking of GH kinematics which utilizes segmentation maps and semantic key points predicted by CNNs.

84]. Iterative methods for solving PnP problems minimize a scalar objective function with numerical optimization and typically include an unknown rotation matrix as a design variable. The constraints which define the Special Orthogonal group (i.e. rotation matrices) result in a non-convex optimization problem and may cause standard non-linear solvers to converge to undesirable local minima, thus returning incorrect pose solutions.

Strategies for handling rotation design variables include global optimization [85–87], convex relaxations [88–93], dual formulations [94, 95], and algebraic methods such as Gröbner bases [96–98]. A particular convex relaxation strategy adapts sums of squares techniques or the closely related characterization of valid moments to globally solve non-convex problems as a series of semidefinite programs (SDPs) [99–105]. In the context of geometric perception, this approach was employed to obtain PnP solutions from a single viewpoint [106], and was extended to incorporate category-level object variability [107,108]. The current work obtains efficient, preliminary pose estimates using 3D model landmarks and 2D key points as part of a multi-stage tracking framework. To accomplish this, it is shown how a multiview PnP formulation can be solved globally using a semidefinite relaxation strategy.

3. Approach

3.1. Overview

The current work investigates a multi-stage framework for automatic tracking of native GH kinematics in stereo-radiography frames (Fig. 1). A stereo-radiography setup consists of two calibrated radiograph systems which capture *in vivo* joint movement as a series of biplanar frames. GH kinematics are obtained by estimating 6-DoF scapula and humerus poses with respect to predefined anatomical coordinate systems, and then aggregating bone poses to obtain joint-level transforms. The problem setup also assumes availability of segmented, subject-specific CT scans, which are used to extract triangular surface meshes and to generate DRRs. The proposed framework consists of three computational steps including preliminary radiograph annotation with CNNs, initial pose estimation from semantic key points, and pose refinement with CT-based DRRs.

3.2. Radiograph annotation with CNNs

A CNN is applied to all radiographs in a captured trial to predict segmentation maps and semantic key points for the scapula and humerus (Fig. 2). Architectures are developed to accept grayscale images as input, $\mathbf{I} \in \mathbb{R}^{h \times w}$, and return tensors representing segmentation maps, $\hat{\mathbf{Y}}_{seg} \in [0,1]^{h \times w \times 2}$, and semantic key point coordinates, $\hat{\mathbf{Y}}_{kp} \in [0,1]^{(2\times 2)K}$, where K is the number of key points assigned to each geometry. Two distinct CNN architectures are investigated.

The first CNN is an encoder–decoder [56] architecture, where the encoder is a ResNet-18 [65] pretrained on ImageNet [109]. An initial convolutional layer projects grayscale images to three channels prior to application of the ResNet encoder. The key point prediction module is a regression layer located at the spatial bottleneck of the network. The final convolutional layer of the ResNet encoder is reshaped and fed to two fully connected layers. The fully connected output is passed through a sigmoid activation function to normalize outputs to the range $[0,1]$. This key point prediction module returns a vector with dimension $4K$ that encodes normalized 2D key point locations for the scapula and humerus ((s) and (h)):

$$\hat{\mathbf{Y}}_{\text{kp}} = [\hat{x}_1^{(s)}, \hat{y}_1^{(s)}, \ldots, \hat{x}_K^{(s)}, \hat{y}_K^{(s)}, \hat{x}_1^{(h)}, \hat{y}_1^{(h)}, \ldots, \hat{x}_K^{(h)}, \hat{y}_K^{(h)}]^T \tag{1}$$

Each element in the prediction vector estimates the x- or y-coordinate of a predefined key point. The architecture's decoder produces segmentation predictions and is augmented with residual-style (instead of concatenation) long range skip connections between encoder and decoder layers with compatible dimensions. Segmentation predictions are obtained from the final convolutional layer in the decoder, which is augmented with a sigmoid activation. Use of a sigmoid rather than softmax activation is critical as a given radiograph pixel may belong to multiple geometries simultaneously. This architecture is referred to as the baseline CNN.

A second CNN extends the first architecture by exploiting subject-specific geometric information in the form of surface meshes. A key observation is that a CNN will predict highly accurate segmentation maps when *ground truth* segmentation is available as input. In the absence of ground truth segmentation at inference time, it may be possible to provide approximate ground truth segmentation in the form of projection maps obtained from PnP solutions. Therefore, the second architecture incorporates a module that uses PnP solutions to obtain projection maps of subject-specific bones, which are then provided as supplementary context to inform segmentation predictions. Key point predictions are obtained from a ResNet encoder (referred to as Encoder 1) similarly to the baseline CNN. Predicted key points and previously obtained bone surface meshes are then used as input data to solve a PnP problem. Surface meshes are subjected to optimized poses and

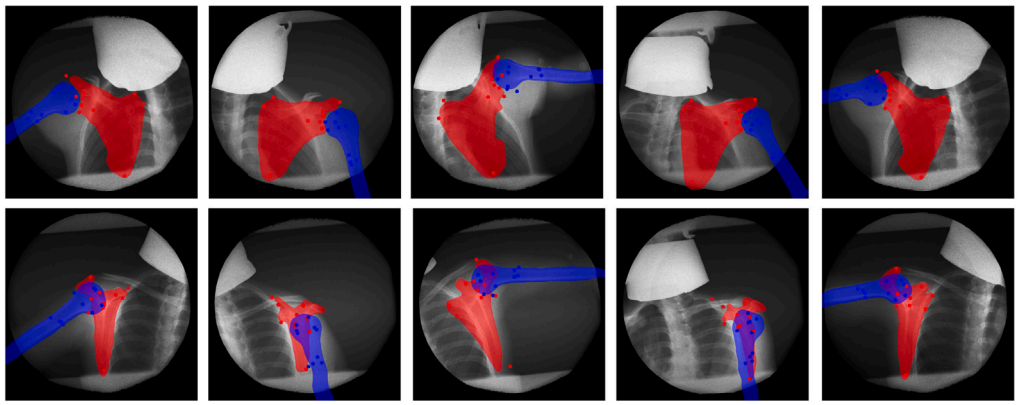

Fig. 2. Radiographs with CNN-predicted segmentation maps and semantic key points for the scapula and humerus. Each column illustrates two different viewpoints from the same biplanar scene. Note that radiographs are contrast-enhanced for illustration purposes.

projected onto the image plane of a virtual camera. The resulting projection map is passed through a second ResNet encoder (Encoder 2) and concatenated with a tensor from the Encoder 1 bottleneck. Fused context from both encoders is then fed to a standard decoder for predicting segmentation maps. The PnP solver is embedded in the network as a backpropagatable module [76,77], and projection maps based on optimized poses are generated with a differentiable renderer [79]. These differentiable modules result in a fully backpropagatable architecture that can be trained end-to-end. This architecture is referred to as the *model-based segmentation* (MBS) CNN, and is developed to support each bone separately.

There are multiple options for providing supervision to the MBS CNN. In addition to directly supervising segmentation ($L(\hat{\mathbf{Y}}_{seg}, \mathbf{Y}_{seg})$) and key point predictions ($L(\hat{\mathbf{Y}}_{kp}, \mathbf{Y}_{kp})$), it is also possible to provide supervision at different stages involved in projection map generation. Three additional supervision components are proposed. First, PnP-optimized poses can be supervised using similar poses obtained from ground truth key points ($L(\hat{\mathbf{P}}, \mathbf{P})$). Next, projection maps obtained from the differentiable rendering module can be supervised with ground truth segmentation ($L(\hat{\mathbf{Y}}_{proj}, \mathbf{Y}_{seg})$). Finally, key point predictions can be supervised using projected model points from PnP solutions ($L(\hat{\mathbf{Y}}_{kp}, \mathbf{Y}'_{kp-proj})$). Therefore, the composite loss function for training the MBS CNN is:

$$\begin{aligned}\lambda_0 L(\hat{\mathbf{Y}}_{seg}, \mathbf{Y}_{seg}) + \lambda_1 L(\hat{\mathbf{Y}}_{kp}, \mathbf{Y}_{kp}) + \lambda_2 L(\hat{\mathbf{P}}, \mathbf{P}) + \\ \lambda_3 L(\hat{\mathbf{Y}}_{proj}, \mathbf{Y}_{seg}) + \lambda_4 L(\hat{\mathbf{Y}}_{kp}, \mathbf{Y}'_{kp-proj})\end{aligned} \tag{2}$$

The hyperparameters $\lambda_{0:4}$ control contributions of each term. Loss components related to key points and poses are instantiated as mean-squared error while binary cross-entropy is used for segmentation loss components.

3.3. Initial pose estimation with biplanar PnP

3.3.1. Problem formulation

Initial pose estimates are obtained for the scapula and humerus using registered, bone-specific surface meshes and 2D key point predictions from both radiographs in a biplanar frame. Although the MBS CNN can provide initial pose solutions from the embedded PnP module, such estimates are obtained based on key points from only a single image. In contrast, a biplanar setup enables use of semantic key points from multiple views which can lead to more accurate pose estimates.

Suppose that the ith 3D model point of a target object is observed by camera c and is projected onto the image plane based on the full-perspective camera model. This 2D point, represented in pixel coordinates $(x_i^{(c)}, y_i^{(c)})$, corresponds to a ray $\mathbf{v}_i^{(c)} \in \mathbb{R}^{3\times1}$ which is back-projected from the image plane and extends through the local camera frame:

$$\mathbf{v}_i^{(c)} = \begin{bmatrix} (x_i^{(c)} - c_x^{(c)})/f_x^{(c)} \\ (y_i^{(c)} - c_y^{(c)})/f_y^{(c)} \\ 1 \end{bmatrix} \tag{3}$$

The camera intrinsic parameters, which include the focal length (f_x, f_y) and principal point (c_x, c_y), are obtained from camera calibration. A 3D point in the local camera frame can be projected onto this ray through application of an orthogonal projection operator:

$$\mathbf{V}_i^{(c)} = \frac{\mathbf{v}_i^{(c)}\mathbf{v}_i^{(c)T}}{\mathbf{v}_i^{(c)T}\mathbf{v}_i^{(c)}} \tag{4}$$

When the projection operator is applied to a point $\mathbf{b} \in \mathbb{R}^{3\times1}$ which is already intersected by the ray, then the result is the original coordinates of the point in question:

$$\mathbf{V}_i^{(c)}\mathbf{b} = \mathbf{I}_3\mathbf{b} \tag{5}$$

where $\mathbf{I}_3 \in \mathbb{R}^{3\times3}$ is the identity matrix. This insight can be used to estimate an object's pose in the local camera frame from pairs of 2D key points and 3D model landmarks. An optimization problem minimizes the sum of squared errors between transformed model points and their projections onto corresponding back-projected rays, constructed from estimated 2D key points [82]:

$$\begin{aligned}&\underset{\mathbf{R},\mathbf{t}}{\text{minimize}} \sum_{i=1}^{K} \|(\mathbf{I}_3 - \hat{\mathbf{V}}_i^{(c)})(\mathbf{R}\mathbf{b}_i^{(c)} + \mathbf{t})\|_2^2 \\ &\text{subject to } \mathbf{R} \in \mathrm{SO}(3)\end{aligned} \tag{6}$$

where projection operators $\hat{\mathbf{V}}_i^{(c)} \in \mathbb{R}^{3\times3}$ are constructed from CNN-predicted key points, $\mathbf{t} \in \mathbb{R}^{3\times1}$ is an unknown translation, and $\mathbf{R} \in \mathbb{R}^{3\times3}$ is constrained to the Special Orthogonal group, $\mathrm{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3\times3} \mid \mathbf{R}^T\mathbf{R} = \mathbf{I},\ \det(\mathbf{R}) = 1\}$. The objective can be extended to incorporate key points from two distinct images by introducing camera extrinsics for each viewpoint:

$$\begin{aligned}&\underset{\mathbf{R},\mathbf{t}}{\text{minimize}} \sum_{c=1}^{2}\sum_{i=1}^{K} \|(\mathbf{I}_3 - \hat{\mathbf{V}}_i^{(c)})(\mathbf{R}_c\mathbf{R}\mathbf{b}_i^{(c)} + \mathbf{R}_c\mathbf{t} + \mathbf{t}_c)\|_2^2 \\ &\text{subject to } \mathbf{R} \in \mathrm{SO}(3)\end{aligned} \tag{7}$$

The constants $\mathbf{R}_c \in \mathbb{R}^{3\times3}$ and $\mathbf{t}_c \in \mathbb{R}^{3\times1}$ specify a calibrated transform from a predefined world frame to camera c's coordinate system. Constraints on $\mathbf{R}$ can be characterized by 15 scalar equalities (or 30 non-strict inequalities) [95,107,110]:

$$\begin{aligned}&1 - \|\mathbf{r}_x\| = 0, && x \in \{0,1,2\} \\ &\mathbf{r}_x^T\mathbf{r}_y = 0, && (x,y) \in \{(0,1),(0,2),(1,2)\} \\ &\mathbf{r}_x \times \mathbf{r}_y - \mathbf{r}_z = \mathbf{0}, && (x,y,z) \in \{(0,1,2),(1,2,0),(2,0,1)\}\end{aligned} \tag{8}$$

The vectors $\mathbf{r}_{(.)} \in \mathbb{R}^{3\times 1}$ are columns of $\mathbf{R}$, and "×" specifies a vector cross product. Note that the first two sets of constraints are scalar-valued, while constraints from the last set are vector-valued and unpacked to recover scalar equalities. Arbitrary bounds can also be placed on $\mathbf{t}$ to restrict the feasible set to a compact basic semialgebraic set. The optimization problem (7) is non-convex due to the constraints on $\mathbf{R}$, and is therefore difficult to solve reliably with standard non-linear solvers.

3.3.2. Solution method with semidefinite relaxations

The biplanar PnP problem (7) is represented as a polynomial optimization problem (POP) and solved using a semidefinite relaxation approach. Let f and g_j be scalar-valued polynomials. In particular, $f = \mathbf{p}^T[\mathbf{x}]_t^N$, where $[\mathbf{x}]_t^N$ is the canonical monomial basis for the polynomials in N variables with degree at most t, and $\mathbf{p}$ is a vector of real-valued coefficients. Also let $\mathbf{x}^{\boldsymbol{\alpha}}$ denote a monomial in the basis $[\mathbf{x}]_t^N$, where $\boldsymbol{\alpha}$ is an N-tuple of non-negative integers with $|\boldsymbol{\alpha}| \le t$. For example, $[\mathbf{x}]_3^2 = [1, x_1, x_2, x_1^2, x_1x_2, \dots, x_1x_2^2, x_2^3]^T$ with dimension $\binom{N+t}{t} = 10$. The degree of f, denoted by $\deg(f)$, is the degree of the highest-degree monomial in f, and similarly for $\deg(g_j)$. A POP is defined as [101–103,111]:

$$\begin{aligned} &\underset{\mathbf{x}}{\text{minimize}}\ f \\ &\text{subject to } g_j \ge 0, \qquad j = 1, \dots, n \end{aligned} \tag{9}$$

where the dependence on $\mathbf{x}$ is implicit. POPs are non-convex in general and include Eq. (7) as a particular instance. Specifically, the objective in Eq. (7) is a degree-2 polynomial with 12 scalar variables comprising 91 distinct monomials, while the constraints on $\mathbf{R}$ and $\mathbf{t}$ are quadratic and linear, respectively.

A method known as *Lasserre's hierarchy* of moment relaxations [101,102,111] proposes to solve Eq. (9) by introducing a series of semidefinite relaxations whose solutions converge to a tight lower bound on the POP for a wide range of problem instances. Broadly, the method involves replacing monomial terms with *lifted design variables* and introducing linear and semidefinite constraints.

Define the *moment matrix* of order k to be the matrix constructed by lifting elements of a vector outer product between identical monomial bases. That is, $\mathbf{M}_k^N(\mathbf{y}) = L_{\mathbf{y}}([\mathbf{x}]_k^N([\mathbf{x}]_k^N)^T) \in \mathbb{R}^{\binom{N+k}{k}\times\binom{N+k}{k}}$, where $L_{\mathbf{y}}(\mathbf{x}^{\boldsymbol{\alpha}}) = y_{\boldsymbol{\alpha}}$ is applied element-wise and replaces each monomial element $\mathbf{x}^{\boldsymbol{\alpha}}$ from the outer product result with an introduced lifted variable $y_{\boldsymbol{\alpha}}$. For example, $\mathbf{M}_2^2(\mathbf{y})$ is the following (also see e.g. [100–102,112,113]):

$$\begin{aligned} \mathbf{M}_2^2(\mathbf{y}) &= L_{\mathbf{y}}\begin{bmatrix} 1 & x_1 & x_2 & x_1^2 & x_1x_2 & x_2^2 \\ x_1 & x_1^2 & x_1x_2 & x_1^3 & x_1^2x_2 & x_1x_2^2 \\ x_2 & x_1x_2 & x_2^2 & x_1^2x_2 & x_1x_2^2 & x_2^3 \\ x_1^2 & x_1^3 & x_1^2x_2 & x_1^4 & x_1^3x_2 & x_1^2x_2^2 \\ x_1x_2 & x_1^2x_2 & x_1x_2^2 & x_1^3x_2 & x_1^2x_2^2 & x_1x_2^3 \\ x_2^2 & x_1x_2^2 & x_2^3 & x_1^2x_2^2 & x_1x_2^3 & x_2^4 \end{bmatrix} \\ &= \begin{bmatrix} y_{00} & y_{10} & y_{01} & y_{20} & y_{11} & y_{02} \\ y_{10} & y_{20} & y_{11} & y_{30} & y_{21} & y_{12} \\ y_{01} & y_{11} & y_{02} & y_{21} & y_{12} & y_{03} \\ y_{20} & y_{30} & y_{21} & y_{40} & y_{31} & y_{22} \\ y_{11} & y_{21} & y_{12} & y_{31} & y_{22} & y_{13} \\ y_{02} & y_{12} & y_{03} & y_{22} & y_{13} & y_{04} \end{bmatrix} \end{aligned} \tag{10}$$

Additionally, define the *localizing matrix* of order k associated with polynomial g to be a similar matrix $\mathbf{M}_k^N(g\mathbf{y}) = L_{\mathbf{y}}(g[\mathbf{x}]_k^N([\mathbf{x}]_k^N)^T) \in \mathbb{R}^{\binom{N+k}{k}\times\binom{N+k}{k}}$, where now each element of the monomial basis outer product incorporates the polynomial g prior to lifting. For example, $\mathbf{M}_1^2(g\mathbf{y})$ with $g = 1 + x_1 + x_2$ is the following:

$$\begin{aligned} \mathbf{M}_1^2(g\mathbf{y}) &= L_{\mathbf{y}}\begin{bmatrix} g & gx_1 & gx_2 \\ gx_1 & gx_1^2 & gx_1x_2 \\ gx_2 & gx_1x_2 & gx_2^2 \end{bmatrix} \\ &= \begin{bmatrix} y_{00}+y_{10}+y_{01} & y_{10}+y_{20}+y_{11} & y_{01}+y_{11}+y_{02} \\ y_{10}+y_{20}+y_{11} & y_{20}+y_{30}+y_{21} & y_{11}+y_{21}+y_{12} \\ y_{01}+y_{11}+y_{02} & y_{11}+y_{21}+y_{12} & y_{20}+y_{12}+y_{03} \end{bmatrix} \end{aligned} \tag{11}$$

Let $d_f = \lceil \deg(f)/2 \rceil$, $d_{g_j} = \lceil \deg(g_j)/2 \rceil$, and $d_K = \max(\{d_{g_j}, j = 1, \dots, n\})$. Then for $k \ge \max(\{d_f, d_K\})$, the *moment relaxation* of order k associated with Eq. (9) is:

$$\begin{aligned} &\underset{\mathbf{y}}{\text{minimize}}\ \mathbf{p}^T\mathbf{y} \\ &\text{subject to } y_{00} = 1 \\ &\qquad \mathbf{M}_k^N(\mathbf{y}) \succeq 0 \\ &\qquad \mathbf{M}_{k-d_{g_j}}^N(g_j\mathbf{y}) \succeq 0, \qquad j = 1, \dots, n \end{aligned} \tag{12}$$

where the optimization is now performed over the vector of lifted variables $\mathbf{y}$, and the notation $\mathbf{M} \succeq 0$ constrains the moment and localizing matrices to the positive semidefinite (PSD) cone [114]. That is, $\mathbf{M} \in \mathbb{S}^d$ with $\mathbf{v}^T\mathbf{M}\mathbf{v} \ge 0$ for all $\mathbf{v} \in \mathbb{R}^d$, and $\mathbb{S}^d$ is the set of $d \times d$ symmetric matrices. In addition, the localizing matrices are formed such that monomial terms observed prior to lifting have at most degree $2k$.

The resulting convex optimization problem is a relaxation of the POP, and can be shown to be an SDP [103,115,116]. The objective is first represented as the matrix inner product of the moment matrix and an additional symmetric matrix, $\mathbf{p}^T\mathbf{y} = \langle \mathbf{A}_0, \mathbf{M}_k^N(\mathbf{y})\rangle = \mathrm{tr}(\mathbf{A}_0^T\mathbf{M}_k^N(\mathbf{y}))$, for some $\mathbf{A}_0 \in \mathbb{S}^d$. Next, the moment and localizing matrices are aggregated to form a block diagonal PSD design matrix, $\mathbf{X} = \mathrm{diag}(\mathbf{M}_k^N(\mathbf{y}), \mathbf{M}_{k-d_{g_1}}^N(g_1\mathbf{y}), \dots, \mathbf{M}_{k-d_{g_n}}^N(g_n\mathbf{y}))$. Equality constraints of the form $\langle \mathbf{A}_\ell, \mathbf{M}_k^N(\mathbf{y})\rangle = 0$ are introduced to tie like terms in the moment matrix, where $\mathbf{A}_\ell \in \mathbb{S}^d$. A similar constraint restricts the upper left element of the moment matrix to unity. Finally, equality constraints relate moment terms in $\mathbf{M}_k^N(\mathbf{y})$ to localizing matrix expressions in $\mathbf{M}_{k-d_{g_j}}^N(g_j\mathbf{y})$, of the form $\langle \mathbf{A}_\ell, \mathbf{X}\rangle = b_\ell$. This leads to minimization of a linear objective over the intersection of an affine subspace and the cone of PSD matrices:

$$\begin{aligned} &\underset{\mathbf{X}}{\text{minimize}}\ \langle \mathbf{A}_0, \mathbf{X}\rangle \\ &\text{subject to } \langle \mathbf{A}_\ell, \mathbf{X}\rangle = b_\ell, \qquad \ell = 1, \dots, m \\ &\qquad \mathbf{X} \succeq 0 \end{aligned} \tag{13}$$

where each $\mathbf{A}_\ell$ is also symmetric. This is the primal SDP form and the feasible set is called a *spectrahedron* [103].

Assuming feasibility, the relaxation (12) can be solved globally with standard SDP solvers [117–120]. It has also been shown that solving Eq. (12) at iteratively increasing relaxation orders leads to certifiable, tight lower bounds on the original problem for a wide range of POPs [101,102,121]. Specifically, if $\mathrm{rank}(\mathbf{M}_k^N(\mathbf{y}^*)) = \mathrm{rank}(\mathbf{M}_{k-d_K}^N(\mathbf{y}^*))$, where $\mathbf{M}_k^N(\mathbf{y}^*)$ is the optimal moment matrix at relaxation order k, then the optimal value of the relaxation coincides with that of the original POP. If $\mathrm{rank}(\mathbf{M}_k^N(\mathbf{y}^*)) = 1$, then the matrix can be decomposed and the minimizers directly unpacked. That is, $\mathbf{M}_k^N(\mathbf{y}^*) = [\mathbf{x}^*]_k^N([\mathbf{x}^*]_k^N)^T$ and the elements corresponding to degree-1 monomials in the extracted monomial basis are the desired minimizers. Cases in which $\mathrm{rank}(\mathbf{M}_k^N(\mathbf{y}^*)) > 1$ are treated separately [122].

In summary, initial pose estimates are obtained for the scapula and humerus by registering 3D model landmarks to 2D key points from two distinct radiographs. A biplanar PnP problem is solved by relaxing the POP representation to a series of convex relaxations in the form of SDPs, which empirically and certifiably converge to a tight lower bound on the original problem. Crucially, global minimizers for the original POP can be acquired from SDP solutions at a given relaxation order k, contingent on a rank condition on the induced moment matrix. Application of Lasserre's hierarchy therefore proceeds by solving Eq. (12) at increasing relaxation orders until either the rank condition is achieved, or a predetermined timeout is reached.

3.4. Pose refinement with model-image registration

Initial pose estimates are refined using model-image registration (Fig. 3). A derivative-free global optimization method searches for

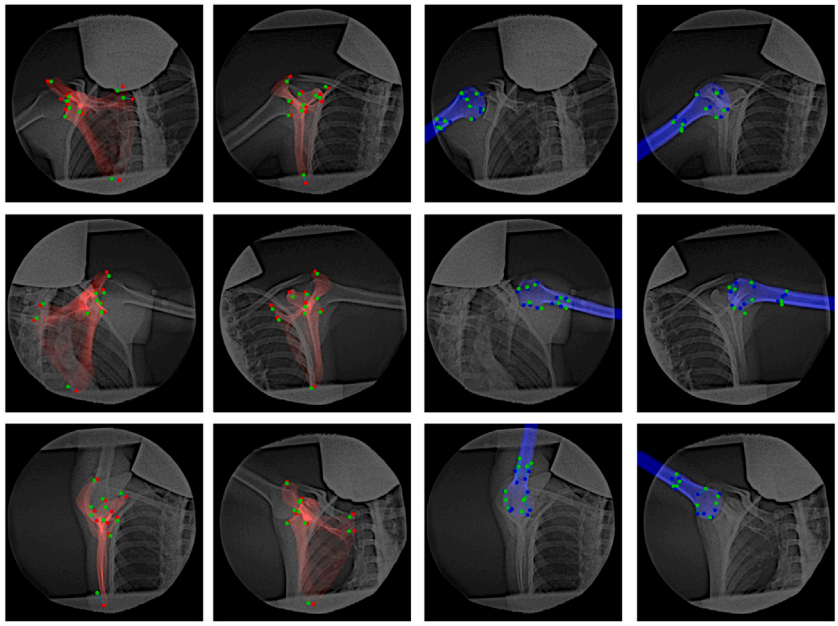

Fig. 3. Scapula and humerus poses estimated by the pose refinement step. Each row shows both views for a single biplanar frame with reference key points utilized in the regularization term (green) and corresponding reprojected model landmarks (red and blue). Note that radiographs are contrast-enhanced for illustration purposes.

refined poses to maximize an objective function composed of two distinct terms. The first term is an intensity-based, patch-wise *similarity metric* between dynamically generated DRRs and radiographs which have been *masked* by CNN-predicted segmentation maps. The second term is a *regularization* component that penalizes differences between CNN-predicted key points and projected 3D model landmarks [42].

Suppose $\widehat{\mathbf{I}}^{(c)} \in [0,1]^{h \times w}$ is the cth masked radiograph from the biplanar setup ($c \in \{1,2\}$), and $\widehat{\mathbf{Y}}^{(c)}_{kp} \in [0,1]^{K \times 2}$ is the set of corresponding key point predictions. Also, let $\mathbf{D}'^{(c)}(\mathbf{p}) \in [0,1]^{h \times w}$ and $\mathbf{Y}'^{(c)}_{kp-proj}(\mathbf{p}) \in [0,1]^{K \times 2}$ represent a rendered DRR and reprojected bone landmarks at a given query pose $\mathbf{p}$. The pose refinement step maximizes:

$$J(\mathbf{p}) = \sum_{c=1}^{2} \sum_{\substack{m,n \\ =1}}^{h,w} \frac{\mathbb{1}(\hat{i}^{(c)}_{mn} > 0)}{h \times w} \sum_{\substack{r= \\ m-10}}^{m+10} \sum_{\substack{f= \\ n-10}}^{n+10} \frac{(\hat{i}^{(c)}_{rf} - \mu^{(c)}_{i,mn})(d'^{(c)}_{rf} - \mu^{(c)}_{d,mn})}{\sigma^{(c)}_{i,mn}\sigma^{(c)}_{d,mn}} - \frac{\lambda_{\text{reg}}}{K} \| \widehat{\mathbf{Y}}^{(c)}_{kp} - \mathbf{Y}'^{(c)}_{kp-proj} \|_F \tag{14}$$

where $\|.\|_F$ is the Frobenius norm, λ_{reg} is a weighting factor, $\hat{i}^{(c)}$ and $d'^{(c)}$ are pixel-specific intensity values from $\widehat{\mathbf{I}}^{(c)}$ and $\mathbf{D}'^{(c)}$, and the indicator function $\mathbb{1}(.)$ evaluates to one if its argument is true, and zero otherwise. The first term specifies that only radiograph pixels which have been retained by segmentation maps should contribute to the similarity metric. For pixel locations where this condition holds, a normalized correlation is computed between DRR and radiograph patches, where $\mu_{.,mn}$ and $\sigma_{.,mn}$ are local means and standard deviations computed from intensity values in patches centered at (m,n) [123].

Similarity metric evaluation is accelerated using a strategy called *dynamic map cropping*, in which the bounds of predicted segmentation maps constrain summations over patches. Suppose the spatial extent of a segmentation map in image c is characterized by $x^{(c)}_{\min}$, $x^{(c)}_{\max}$, $y^{(c)}_{\min}$, and $y^{(c)}_{\max}$, and let $\lambda \in [0,1]$ represent a hyperparameter. Then dynamic map cropping reduces the set of patch candidates evaluated in the similarity metric.

$$J'(\mathbf{p}) = \sum_{c=1}^{2} \sum_{\substack{m= \\ y^{(c)}_{\min} - \lambda h}}^{y^{(c)}_{\max} + \lambda h} \sum_{\substack{n= \\ x^{(c)}_{\min} - \lambda w}}^{x^{(c)}_{\max} + \lambda w} \frac{\mathbb{1}(\hat{i}^{(c)}_{mn} > 0)}{h \times w} \sum_{\substack{r= \\ m-10}}^{m+10} \sum_{\substack{f= \\ n-10}}^{n+10} \frac{(\hat{i}^{(c)}_{rf} - \mu^{(c)}_{i,mn})(d'^{(c)}_{rf} - \mu^{(c)}_{d,mn})}{\sigma^{(c)}_{i,mn}\sigma^{(c)}_{d,mn}} \tag{15}$$

Dynamic map cropping improves the speed and storage requirements for a single objective function call. Reducing the set of evaluated pixels enables the DRR-generation subroutine to trace a reduced number of virtual rays through the CT volume using Siddon's or related methods [124,125]. Additionally, limiting the summation bounds over patch centers leads to faster aggregation of patch-wise similarity contributions, even when exploiting parallel reduction methods, as is done in the current investigation.

In practice, poses $\mathbf{p} \in \mathbb{R}^{6 \times 1}$ are parameterized as ZYX Euler angles and translation vectors. Values are sampled from a 6-dimensional Gaussian distribution with mean vector $\boldsymbol{\mu}_t \in \mathbb{R}^{6 \times 1}$ and covariance matrix $\boldsymbol{\Sigma}_t \in \mathbb{S}^6$, controlled using CMA-ES [42,44,126]. The P poses corresponding to the top 50% of objective values at each time step t are used to update the mean and covariance.

$$\boldsymbol{\mu}_{t+1} = \lambda_\mu \boldsymbol{\mu}_t + \frac{(1-\lambda_\mu)}{P} \sum_{j=1}^{P} \mathbf{p}_{j,t} \tag{16}$$

$$\boldsymbol{\Sigma}_{t+1} = \lambda_\Sigma \boldsymbol{\Sigma}_t + \frac{(1-\lambda_\Sigma)}{P} \sum_{j=1}^{P} (\mathbf{p}_{j,t} - \boldsymbol{\mu}_t)(\mathbf{p}_{j,t} - \boldsymbol{\mu}_t)^T \tag{17}$$

The hyperparameters λ_μ and λ_Σ control how aggressively the sampling distribution is updated. CMA-ES is applied for a fixed number of time steps at multiple scales, or *outer-loops*, using distinct radiograph and CT resolutions. The mean and covariance are initialized at the first scale with predefined values. The mean vector is carried over at each new scale while the covariance matrix is reset. Pose refinement is performed for the scapula and humerus separately using initial pose estimates, CNN predictions, radiographs, DRRs, and surface meshes.

4. Methods

4.1. Data

The automatic tracking method was evaluated using stereo-radiography trials and manually tracked GH kinematics previously collected for a population-based study [35]. Ten subjects with no known history of upper-extremity injuries performed dynamic activities with left and right shoulders in front of an HSSR system [41]. Images were captured at 25 frames per second with dimensions of 1024 × 1024 pixels. A

total of 86 trials were captured with an average of 93 frames per sequence. Activities performed by subjects included arm abduction, forward flexion, and adducted and abducted internal–external rotation.

Bilateral CT scans were obtained for all subjects and segmented using commercial software (ScanIP, Simpleware, Exeter, UK) to isolate the scapula and humerus. Segmented bones were reconstructed into surface meshes and represented in anatomical coordinate systems [16]. Surface meshes were registered to bone-specific template meshes with a custom workflow involving rigid [127] and non-rigid [128] point cloud registration. Ten landmarks were manually annotated on scapula and humerus template meshes. Scapula landmarks included 5 locations on the glenoid, termini of the acromion and coracoid process, and 3 points on the plate border. Humerus landmarks included 6 locations around the humeral head and 4 locations on the surgical neck. Precomputed data structures were also created from CT scans for DRR generation by storing only voxels within volumetric bounding boxes encapsulating segmented bones.

GH kinematics were tracked with respect to anatomical coordinate systems using commercial software (DSX, C-Motion Inc., Germantown, MD, USA). Trials were processed by manually tracking every tenth frame and invoking built-in optimization to track remaining frames. Results were visually inspected and corrected if needed. Radiographs were associated with ground truth segmentation and key point annotations by projecting tracked, registered surface meshes onto the image plane. Key points were defined based on the preselected scapula and humerus landmarks. Custom MATLAB code (Mathworks, Natick, MA) was created to compute joint-level transforms from individual bone poses as $\mathbf{T}_{\mathrm{gh}} = \mathbf{T}_{\mathrm{s}}^{-1}\mathbf{T}_{\mathrm{h}}$, where $\mathbf{T}_{\mathrm{s}} \in \mathbb{R}^{4\times4}$ and $\mathbf{T}_{\mathrm{h}} \in \mathbb{R}^{4\times4}$ are tracked poses of the scapula and humerus in homogeneous transform format with respect to anatomical coordinate systems [35,129].

Annotated radiograph trials were partitioned into training, validation, and test sets for development and evaluation of the automatic tracking framework. The training set included 67 trials from 8 subjects. The validation set included two additional trials performed by subjects that were represented in the training set. The test set consisted of 17 trials representing both left and right shoulders and all 4 activities, performed by two individuals not present in the training set. The training, validation, and test partitions were composed of 5816, 216, and 1973 frames, respectively, where each frame included 2 images from the biplanar setup.

4.2. Experiments

The automatic tracking method was developed and evaluated with the partitioned data set. Experiments were performed to answer the following questions: (1) Does the proposed framework enable fully automatic tracking of GH kinematics from stereo-radiography? (2) Does MBS improve the accuracy of predicted segmentation maps in radiograph scenes? and (3) Does segmentation accuracy affect downstream pose estimates?

Baseline and MBS CNN models were developed with ground truth segmentation and key point annotations from the training set. CNNs were trained for 500,000 iterations using Adam optimization [130] with a batch size of 1 and learning rate of 5×10^{-6}. Standard data augmentation was performed including rotation, translation, brightness adjustment, and blurring. Eq. (2) was used for training with $\lambda_0 = 1$ and $\lambda_1 = 10$ for both CNN architectures. The baseline CNN was trained with $\lambda_{2:4} = 0$ as these loss components were not applicable to the vanilla encoder–decoder architecture. Images were resized to 256×256 pixels for training and inference.

The MBS CNN was implemented using distinct strategies for fusing context from encoded radiographs (Encoder 1) and encoded projection maps from the embedded PnP module (Encoder 2). Tensors obtained from the spatial bottlenecks of Encoder 1 and Encoder 2 were nominally fused via concatenation before being fed to a single decoder. Long-range skip connections were then incorporated into the decoder using tensors from only Encoder 1 (Decoder Style 1), from only Encoder 2 (Decoder Style 2), or from both encoders (Decoder Style 3). Decoder Style 3 was instantiated by concatenating tensors from both encoders with similar spatial dimensions, and then down-sampling the channel dimension before fusing with corresponding decoder scales. These models were trained with $\lambda_{2:4} = 0$.

Experiments were performed to evaluate sensitivity of MBS CNN performance to multiple training conditions. Decoder Style 3 was retrained after disconnecting gradient flow at the embedded PnP and differentiable rendering modules (cut grads) to investigate if end-to-end differentiability affects CNN performance. Decoder Style 3 was also retrained with various weight combinations for supplementary loss components to investigate effects of different supervision styles on CNN performance.

Trained CNNs were applied to radiographs from both viewpoints across all 17 test trials and evaluated against ground truth annotations. Segmentation performance was assessed with Dice Similarity Coefficient (DSC) and predicted key points were compared against ground truth values in pixel units. All CNNs were implemented in PyTorch [131]. MBS CNN architectures were implemented using differentiable rendering functionality from PyTorch3D [79] and the differentiable PnP module from prior work [76].

Initial bone pose estimates were obtained for all test frames by solving the biplanar PnP formulation. CNN-predicted key points from the baseline CNN and Decoder Style 3 MBS CNN were used as problem data. The optimization problem (7) with rotation and translation constraints was solved using the described semidefinite relaxation approach with YALMIP [117,119]. Bone pose estimates were obtained for the scapula and humerus separately.

Pose refinement was implemented by initializing CMA-ES search distributions with initial pose estimates. Multiple experiments were performed in which masked radiographs were constructed from different segmentation map cohorts. Masked radiographs were constructed from baseline CNN segmentation maps, MBS maps, MBS maps dilated with a kernel size of 15 pixels, or ground truth segmentation. Pose refinement was also implemented using original, unmasked radiographs (no mask). Experiments performed with MBS CNN maps were initialized with preliminary poses obtained from MBS CNN-predicted key points. The remaining experiments utilized preliminary pose estimates obtained from baseline CNN key points. Joint-level transforms were recovered from individual bone pose estimates in all experiments.

The CMA-ES sampling distribution was initialized with a mean vector specified by initial pose estimates and an identity covariance matrix. Pose refinement was performed at 4 different scales characterized by different radiograph and DRR resolutions. The first scale consisted of 100 iterations of populations with 64 poses each. Radiographs were downsized to 128×128 pixels and DRRs were generated from CT scans which were downsized to $\frac{1}{4}$, $\frac{1}{4}$, and $\frac{1}{8}$ of their original resolutions in the two in-plane directions and along the inter-slice (superior–inferior) axis, respectively. Let this configuration be denoted by $\{100, 64, 128, [\frac{1}{4}, \frac{1}{4}, \frac{1}{8}]\}$. The second CMA-ES outer loop configuration was $\{200, 32, 256, [\frac{1}{2}, \frac{1}{2}, \frac{1}{4}]\}$. The third and fourth configurations were $\{200, 8, 512, [1, 1, \frac{1}{2}]\}$ and $\{100, 8, 1024, [1, 1, 1]\}$. Values of 0.7 and 0.9 were assigned to λ_μ and λ_Σ. The regularization term in Eq. (14) was implemented with λ_{reg} values of 0, 10, 1E2, 1E3, and 1E4.

Evaluation of Eq. (14) was implemented in pyCUDA [132] and all poses in a given CMA-ES population were evaluated in parallel. Multiple subroutines within each pose evaluation were further parallelized such as matrix multiplication, reduction of patch-wise similarity metric contributions, and DRR pixel computation using the Siddon–Jacobs method [124,125].

Initial and refined pose predictions were evaluated by comparing individual bone poses to manually tracked values. Bone rotation differences were assessed using geodesic distance defined as $|\arccos((\mathrm{trace}(\hat{\mathbf{R}}^T\mathbf{R}) - 1)/2)|$, where $\hat{\mathbf{R}}$ and $\mathbf{R}$ are predicted and ground

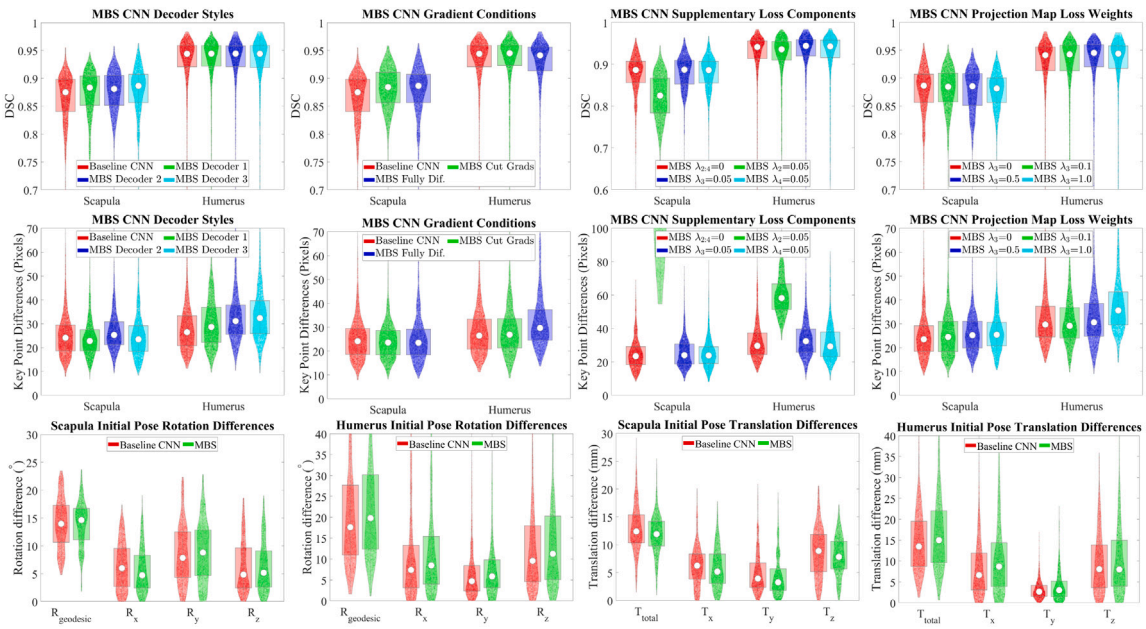

Fig. 4. Performance of baseline and MBS CNNs trained to predict segmentation maps and semantic key points for the scapula and humerus (top two rows), and individual bone pose accuracy achieved by initial pose estimation based on key points from baseline and MBS CNNs (bottom row).

truth rotations, respectively [133,134]. Optimal rotations [135] between predicted and ground truth poses were transformed into bone anatomical coordinate systems and decomposed into ZYX Euler angles. Translation differences were similarly decomposed in the anatomical coordinate system for the scapula and humerus.

Residual transforms between predicted and ground truth joint-level kinematics were computed [136]. Residual transforms were defined as $\mathbf{T}_{\mathrm{r}} = \hat{\mathbf{T}}_{\mathrm{gh}}^{-1}\mathbf{T}_{\mathrm{gh}}$, where $\hat{\mathbf{T}}_{\mathrm{gh}}$ and $\mathbf{T}_{\mathrm{gh}}$ are joint-level transforms obtained from refined pose predictions and manually tracked poses, respectively. These transforms quantified differences in humerus pose with respect to scapula pose, between predicted and ground truth kinematics. Euler angles and geodesic distances were computed from rotation components of residual transforms in all test frames.

Success rates were defined to evaluate individual bone poses and joint-level kinematics. Success rates were computed as the percentage of test frames in which differences between predicted and ground truth poses were below predefined thresholds. Threshold values included 1.0°, 1.5°, 2.0°, and 3.0°.

5. Results

CNNs produced accurate segmentation and key point predictions in radiographs (Fig. 4) [137]. Mean scapula DSC values were 0.865, 0.873, 0.875, and 0.877 for the baseline CNN and MBS CNNs with Decoder Styles 1, 2, and 3, respectively. Mean humerus DSC values were 0.922, 0.923, 0.919, 0.916 for the baseline CNN and MBS CNNs with Decoder Styles 1, 2, and 3. Mean key point errors for these architectures were 24.8, 23.9, 26.7, and 24.7 pixels for the scapula, and 28.0, 30.4, 33.0, and 32.4 pixels for the humerus.

Scapula segmentation performance improved when incorporating subject-specific geometric information into CNN architectures. MBS CNN scapula segmentation improved over baseline predictions in 63%, 65%, and 68% of test images for Decoder Styles 1, 2, and 3. Similar improvements were not observed for humerus segmentation.

CNN-predicted key points enabled coarse bone pose estimates (Fig. 4). Scapula and humerus poses obtained from baseline CNN key points demonstrated median rotation differences of 13.9° and 17.6° compared to ground truth values; and median translation differences of 12.4 mm and 13.5 mm. Scapula and humerus poses obtained from MBS CNN key points demonstrated median rotation differences of 14.6° and 19.8°, and median translation differences of 11.9 mm and 14.9 mm.

Incorporation of masked radiographs during pose refinement enabled improved scapula and humerus pose estimates (Fig. 5, Table 1). Median scapula rotation differences across all test trials were 3.9°, 1.9°, 1.8°, 1.7°, and 1.4° for no masking and masked radiographs created from baseline CNN segmentation maps, MBS CNN maps, dilated MBS CNN maps, and ground truth segmentation maps. Median humerus rotation differences were 8.3°, 7.3°, 8.1°, 8.6°, and 6.6° for no masking and masked radiographs created from baseline CNN segmentation maps, MBS CNN maps, dilated MBS CNN maps, and ground truth segmentation maps.

Accuracy of joint kinematics estimates from pose refinement varied by DoF (Fig. 6) but was relatively invariant across multiple regularization weights, except for the maskless condition (Fig. 7). Median joint kinematics rotation differences were 1.4°, 8.4°, and 1.4° for X, Y, and Z Euler angles between ground truth and refined poses obtained with dilated MBS CNN segmentation maps and a regularization weight of 1E3. Success rates for these DoFs were 53.7%, 10.4%, and 54.3% at the 1.5° threshold, and 65.6%, 13.0%, and 63.3% at the 2.0° threshold (Table 2).

6. Discussion

A method for fully automatic tracking of native GH kinematics in stereo-radiography sequences has been proposed. Although 6-DoF pose estimation has been previously investigated [59,138,139], stringent accuracy requirements associated with radiography tracking applications preclude the use of stand-alone ML-based pose estimation. In addition, while semi-automatic tracking has historically been used to refine preliminary pose estimates, such methods stipulate that bone poses should be manually initialized at precisions which exceed the capabilities of standard pose regression or PnP-based frameworks. The current work demonstrates that aggregation of ML annotation, initial pose estimation, and pose refinement with masked radiographs produces accurate GH kinematics estimates across multiple dynamic activities.

Segmentation and key point annotations were obtained with a proprietary CNN which supported subject-specific geometric information and image data as inputs. A key concern with native bone tracking is

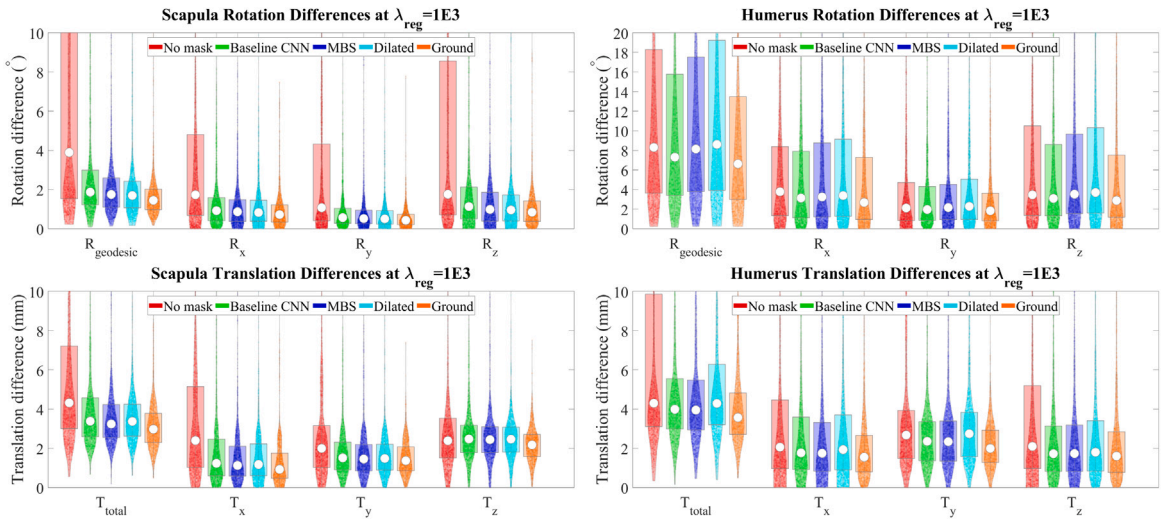

Fig. 5. Individual bone pose differences from pose refinement based on maskless (no mask) or masked radiographs obtained from different segmentation cohorts.

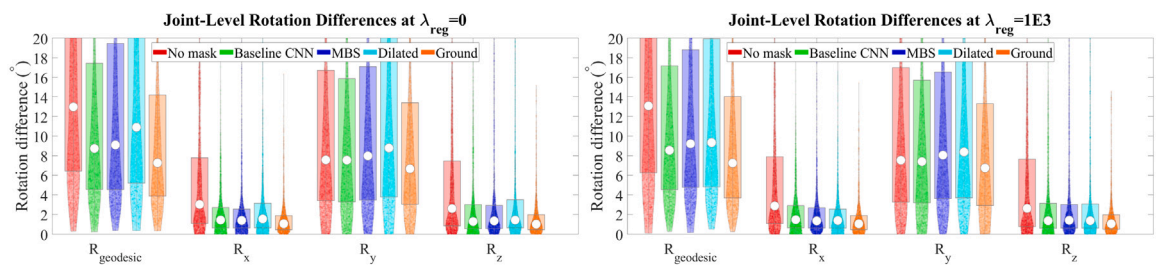

Fig. 6. Joint kinematics differences from pose refinement with different radiograph masking conditions.

Table 1
Bone pose success rates from pose refinement with regularization weight of 1E3 and different radiograph masking conditions.

Geometry	Radiograph masking	Success rate at 1° threshold (%)				Success rate at 1.5° threshold (%)				Success rate at 2° threshold (%)				Success rate at 3° threshold (%)			
		R_{geo}	R_x	R_y	R_z	R_{geo}	R_x	R_y	R_z	R_{geo}	R_x	R_y	R_z	R_{geo}	R_x	R_y	R_z
Scapula	No mask	12.6	35.0	47.9	35.8	24.0	45.7	58.1	47.0	35.0	53.7	63.2	52.5	45.8	62.8	70.2	58.5
	Baseline CNN seg	17.0	52.6	73.3	44.4	36.8	72.8	84.5	61.3	54.2	82.6	90.3	72.3	75.2	89.3	94.9	85.0
	MBS CNN seg	20.5	57.0	76.4	50.7	39.3	75.4	87.4	67.0	59.2	86.0	92.4	77.7	81.9	92.6	95.6	91.4
	Dilated MBS CNN	**21.2**	**59.7**	**78.2**	**51.8**	**41.3**	**76.0**	**90.3**	**69.0**	**61.8**	**86.7**	**94.5**	**81.3**	**83.1**	**92.8**	**96.6**	**90.1**
	Ground truth seg	26.2	65.3	85.7	57.9	52.1	84.1	95.8	77.5	74.0	92.2	98.3	88.7	92.9	96.8	99.6	95.5
Humerus	No mask	3.3	19.8	29.2	20.0	7.4	27.4	39.3	27.2	11.2	33.6	48.2	33.8	19.9	43.0	62.8	45.5
	Baseline CNN seg	3.5	23.0	28.5	19.4	7.7	31.2	41.2	27.8	12.1	38.2	50.6	34.8	21.6	48.9	63.6	49.3
	MBS CNN seg	3.5	22.0	26.5	17.1	6.3	30.8	36.5	24.1	10.2	37.0	47.0	31.0	18.7	47.9	61.3	44.5
	Dilated MBS CNN	**2.9**	**20.3**	**26.2**	**16.5**	**6.7**	**28.7**	**36.0**	**23.8**	**10.0**	**35.2**	**45.0**	**30.0**	**17.8**	**46.0**	**60.3**	**42.9**
	Ground truth seg	4.8	26.2	29.5	22.1	9.8	34.2	43.2	31.0	14.9	41.3	52.9	38.7	25.3	52.8	68.7	51.2

Table 2
Joint kinematics success rates for regularization weight of 1E3 and different radiograph masking conditions.

Radiograph masking	Success rate at 1° threshold (%)				Success rate at 1.5° threshold (%)				Success rate at 2° threshold (%)				Success rate at 3° threshold (%)			
	R_{geo}	R_x	R_y	R_z	R_{geo}	R_x	R_y	R_z	R_{geo}	R_x	R_y	R_z	R_{geo}	R_x	R_y	R_z
No mask	7.9	23.5	8.2	30.0	2.6	31.6	12.3	38.8	4.9	39.3	16.0	43.9	9.5	51.1	23.2	53.3
Baseline CNN seg	1.1	38.3	8.7	39.6	3.5	50.8	12.6	53.7	6.0	61.6	16.6	62.1	13.4	76.5	23.5	73.3
MBS CNN seg	1.2	38.4	6.4	38.2	3.1	54.3	9.5	52.1	5.0	64.0	13.7	62.7	11.4	79.4	20.6	74.7
Dilated MBS CNN	**1.3**	**38.2**	**7.2**	**38.0**	**3.3**	**53.7**	**10.4**	**54.3**	**5.7**	**65.6**	**13.0**	**63.3**	**12.4**	**79.5**	**20.1**	**74.4**
Ground truth seg	2.1	47.4	9.1	48.3	5.5	65.6	13.2	64.1	8.7	77.2	17.2	75.6	18.1	89.2	25.7	86.8

that of low visibility. The scapular plate is typically hidden by adjacent anatomy while the coracoid process and acromion are "washed out" and exhibit poor contrast against surrounding tissue. As a result, it is common that only regions near the glenoid are clearly visible on the scapula. The combination of insufficient radiograph visibility and significant population-level anatomical variation [140] makes accurate contour delineation a challenging problem when relying on image data alone. The MBS architecture in the current work attempted to address

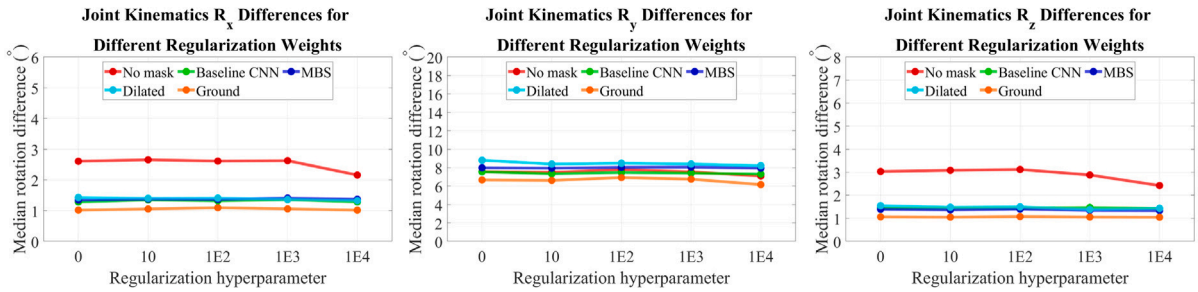

Fig. 7. Joint kinematics differences from pose refinement across multiple regularization weights.

these challenges by informing segmentation predictions with subject-specific projection maps obtained from backpropagatable optimization and differentiable rendering modules. Experiments suggested that the MBS architecture improved predictive performance for scapula segmentation. A drawback of the MBS CNN is that it supported only a single object instance. While single-object support was sufficient for the current problem setup, future work may explore more scalable or improved methods for achieving *model-based inference* in vision architectures.

A biplanar PnP problem was formulated to compute initial pose estimates. An established method, known as Lasserre's hierarchy of moment relaxations [100–102,111], was employed to relax the non-convex, scalar POP to a series of SDPs. This relaxation approach has been instrumental in similar computer vision applications [100,106–108], in which inclusion of rotation design variables leads to challenging non-convex optimization problems [88]. The current work has extended this approach to pose estimation from multiple camera views. A byproduct of this method is the recovery of a sufficient condition which certifies tightness between original POP and relaxed solutions. Such an *a posteriori* solution certificate may contribute a degree of reliability to the fully automatic system [94,110,141]. Alternative solution methods may converge to incorrect local minima or return pose estimates with no guarantee of optimality [52]. In such cases where incorrect results were to occur, unsatisfactory initial pose estimates would go unnoticed in a fully autonomous system and could prevent effective identification of the pose refinement search space. In contrast, solutions to the SDP reformulations can be verified by observing *strong duality*; and solution certificates based on a rank condition on optimal moment matrices verify that relaxed solutions coincide with the original, non-convex POP. A drawback of this relaxation strategy is its poor scalability with respect to the quantity of design variables. Initial pose estimates in the current work were obtained in 10 s per instance using the proposed objective with 12 scalar design variables. Adoption of efficient SDP solvers may accelerate solution times in future work [113,142–144].

Masked radiographs constructed from CNN-predicted segmentation led to improved kinematics predictions during pose refinement. A key concern with conventional semi-automatic tracking is that numerical optimization methods may settle in undesirable local minima due to spurious correlations between DRR and radiograph features [20]. Although this risk is also present for the fully autonomous pipeline, experiments suggested that masking radiographs during pose refinement reduced the frequency of this failure mode. Notably, tracking performance did not vary widely when using baseline CNN, MBS CNN, and dilated MBS CNN segmentation maps, but joint kinematics success rates were highest when masking radiographs with ground truth segmentation. This suggests that more significant improvements in predicted segmentation could further enhance downstream pose estimates by reducing similarity metric contributions from irrelevant radiograph patches.

Kinematic predictions were limited by humerus pose accuracy. While the scapula was characterized by poor visibility, its unique geometric features enabled relatively accurate pose estimates. In contrast, the axisymmetric shape of the humerus affected tracking performance. Rotations about the humeral long axis result in minimal visual changes in radiograph scenes during manual or automatic tracking workflows. The humeral head mitigates total axisymmetry of the proximal humerus, but it is difficult to perceive due to overlap with the glenoid. Humerus visibility was also noted to degrade considerably in views which were precisely orthogonal with the sagittal plane, as this led to overlapping of the humerus with the full torso. These visibility issues impacted success rates, but experiments demonstrated accurate predictions for 2 out of 3 joint-level DoFs. The most significant tracking error was observed in the second DoF, which corresponds to internal–external rotation [16]. The remaining two DoFs, which represented flexion–extension and adduction–abduction, demonstrated considerably higher success rates at all success thresholds. Future tracking studies can mitigate visibility issues by avoiding overlap between the humerus and torso, or avoiding views in which the humeral long axis is perpendicular to the image plane.

The proposed framework builds on recent studies which investigated fully autonomous methods for precise pose estimation in single plane or biplanar radiography. Grupp et al. [42] implemented an automatic system to register the proximal femur and pelvis in single plane radiography. Experiments were performed by training CNNs and estimating poses in a leave-one-out analysis. Bone poses were predicted across 6 trials after manually removing scenes with poor visibility. In addition, ground truth bone poses were computed using an optimization-based approach which was similar to that employed by the evaluated system. In contrast, the current work investigated automatic tracking of GH kinematics in a biplanar setup. The composite system was evaluated across 17 distinct trials capturing a range of dynamic activities, many of which were characterized by poor visibility. Kinematic predictions were compared to manually tracked values previously annotated by trained experts for a population-based study. This assessment across diverse scenes demonstrates the system's robustness in practice.

Burton et al. [52] investigated automatic tracking of healthy knee kinematics in stereo-radiography sequences. Their framework utilized surface mesh-based projection maps rather than DRRs during pose refinement. From a practical standpoint, surface mesh rendering is generally more efficient than DRR generation and enables more pose evaluations for a given computational budget. However, unlike CT-based DRRs, surface mesh projections do not capture cortical bone information which may benefit the pose refinement step. Registration with surface meshes is viable for native bone tracking when paired with accurate segmentation maps, but the presence of degraded segmentation may call for additional visual features provided by DRRs. Preliminary experiments concluded that surface mesh registration was not viable for tracking native GH kinematics.

Recently, Jensen et al. [54] investigated automatic tracking of TKA kinematics in single plane radiography. Preliminary poses were estimated based on NFD libraries, while pose refinement was achieved using surface mesh registration. Notably, their framework demonstrated impressive robustness across multiple implant models and imaging

setups. Considerable radio-opacity of implant components leads to significant contrast against surrounding tissue in radiographs, thus enabling highly accurate CNN inference. In addition, while single plane systems are known to complicate pose estimation for particular DoFs, they also comprise less hardware and are more widely available in research or clinical environments.

There are limitations associated with the current investigation. A drawback of the proposed system is the additional preprocessing required for implementation. Radiograph scenes, surface meshes, and segmented CTs are standard requirements for radiography tracking workflows, but the proposed method also required registering subject-specific bones to predefined template meshes. This registration established correspondence between 3D model landmarks and CNN-predicted key points, which was required for embedded PnP modules in MBS CNNs, initial pose estimation, and the regularization term in Eq. (14). Non-rigid mesh registration is a well-studied problem [128, 145] but presents an additional preprocessing step in the end-to-end tracking workflow. Landmarks on 3D models could alternatively be identified manually. This would result in minimal additional overhead as manual annotation is already required for constructing anatomical coordinate systems. Reliance on subject-specific anatomy could also be circumvented by incorporating statistical shape and intensity models (SSIMs) [140,146], although evidence of improved accuracy would be preferred before introducing more DoFs into the system.

An additional limitation of the proposed method concerns computational efficiency. CNN predictions were obtained in less than 1 s, and initial pose estimates required approximately 10 s per object. However, pose refinement required 1–2 min per geometry on an Nvidia GeForce RTX 2080 Ti. This was due to an aggressive iteration count assigned at each CMA-ES outer loop. While such lead times already improve on currently employed tracking methods, future work will investigate efficiency improvements.

A final consideration regarding performed experiments is that manually tracked kinematics were treated as gold-standard during evaluation. While GH kinematics were annotated by trained experts and are expected to be highly accurate, an alternative evaluation method involves using implanted metallic beads to acquire ground truth poses [22]. Such experiments will be performed in future work to further validate autonomous tracking frameworks.

The current work has presented a framework for fully automatic tracking of native bone kinematics in stereo-radiography sequences. Optimistic success rates for predicted GH kinematics in 2 out of 3 DoFs demonstrate the viability of end-to-end automatic radiography tracking. Fully autonomous systems can accelerate standard practices by orders of magnitude and increase scalability of tracking workflows in research, clinical, and surgical applications.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: William Burton reports equipment, drugs, or supplies was provided by Ortho Haus LLC.

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